



Total Orders
3,467



Total Sales
\$438,097



Top State
California



Top Region
West

Project Overview

This deck documents a regional sales dataset (States by Region) and the Excel formulas used to generate derived columns, plus a concise analytical summary. Audience: business analysts and sales managers. The project calculates order counts, aggregated sales, ranking, segment splits, date-derived metrics, profit slabs, and highlights low-profit values.

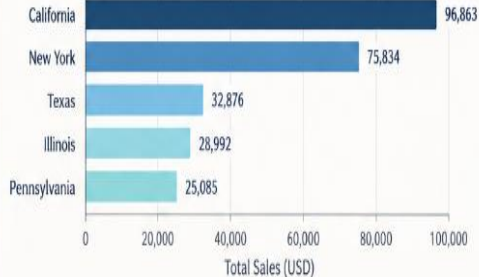
Sales by Region

Total Sales (USD)



Top States by Sales

Total Sales (USD)

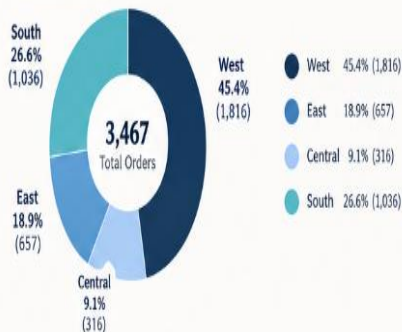


State Performance

State	Region	Total Orders	Total Sales (USD)	Rank
California	West	522	96,863.37	1
New York	East	12	75,833.86	2
Texas	Central	232	32,876.26	3
Washington	West	522	30,219.02	4
Illinois	Central	147	28,991.95	5
Pennsylvania	East	147	25,085.22	6
Michigan	Central	44	20,752.40	7
Florida	South	109	19,823.19	8
Ohio	East	44	19,111.34	9
North Carolina	South	39	17,634.99	10

Orders by Region

Total Orders (3,467)



Data Snapshot

Key Fields



Dataset Highlights

(AGGREGATED ACROSS STATES)



TOTAL CUSTOMERS (ROWS)

20 state entries



GRAND TOTAL SALES

438,097



AGGREGATE METRIC SHOWN

Avg Sale Per Customer
= 37,533 (from input)



PRIMARY SEGMENTS

Corporate, Consumer, Home Office

(Consumer dominates)



What the Project Does



Groups transactional rows by State and Region.



Aggregates Total Orders and Total Sales per state.



Ranks states by sales.



Calculates percent contribution.



Breaks sales into segments (Corporate, Consumer, Home Office).



Extracts first/last order dates.



Computes recency in Years/Months/Days.



Formats last-order month.



Computes average sale, profit, and profit-after-tax.

Mandated formulas — Full list (what each column uses)

Below are the exact formulas supplied for each derived column in the project. Each formula is shown verbatim (Excel syntax) and will be explained in the following slide.

Column B

=VLOOKUP(A2,Sheet1!I2:J2539,2,FALSE)

Column C

=CONCATENATE(A2," : ",B2)

Column D

=COUNTIF(Sheet1!I:I,Sheet1!I2)

Column E

=SUMIF(Sheet1!I:I,A2,Sheet1!P:P)

Column F

=RANK(E2,E\$2:E\$21,0)

Column G

=E2/E\$22

Columns H–J

=SUMIFS(Sheet1!Q:Q,Sheet1!I:I,A2,Sheet1!G:G,\$H\$1) (and similar for \$I\$1,\$J\$1)

Column K

=XLOOKUP(MAX(H2:J2),H2:J2,\$H\$1:\$J\$1)

Columns L–M

=MINIFS(Sheet1!C:C,Sheet1!I:I,A2) and
=MAXIFS(Sheet1!C:C,Sheet1!I:I,A2)

Column N

=DATEDIF(M2,TODAY(),"Y") & " Years " &
DATEDIF(M2,TODAY(),"YM") & " Months " &
DATEDIF(M2,TODAY(),"MD") & " Day "

Column O

=TEXT(M2,"mmm-yy")

Column Q

=AVERAGEIF(Sheet1!I:I,A2,Sheet1!P:P)

Columns R–S

=SUMIF(Sheet1!I:I,A2,Sheet1!S:S)

Column T

=IF(R2<10000,"<10K",IF(AND(R2>=10000,R2<=20000),"10-20K", ">20K"))

Formula explanations — purpose, inputs, and expected result

	A	B	C	D	E	F
1	State	Region	CONCAT (Region : State)	Total Order	Total Sale	Rank By Sale Qty
2	Texas	Central	Texas : Central	232	32876.257	3
3	Illinois	Central	Illinois : Central	147	28991.948	5
4	Michigan	Central	Michigan : Central	44	20752.404	7
5	Indiana	Central	Indiana : Central	522	12086	11
6	Wisconsin	Central	Wisconsin : Central	18	8632.02	14
7	New York	East	New York : East	12	75833.857	2
8	Pennsylvania	East	Pennsylvania : East	147	25085.223	6
9	Ohio	East	Ohio : East	44	19111.337	9
10	Massachusetts	East	Massachusetts : East	25	4566.52	19
11	New Jersey	East	New Jersey : East	47	3638.326	20
12	Florida	South	Florida : South	109	19823.1935	8
13	North Carolina	South	North Carolina : South	39	17634.988	10
14	Virginia	South	Virginia : South	13	10292.75	12
15	Tennessee	South	Tennessee : South	292	5588.371	16
16	Georgia	South	Georgia : South	522	5638.43	15
17	Kentucky	South	Kentucky : South	28	5399.88	18
18	California	West	California : West	522	96863.3655	1
19	Washington	West	Washington : West	522	30219.022	4
20	Arizona	West	Arizona : West	157	9542.167	13
21	Colorado	West	Colorado : West	25	5520.677	17
22	Total			3467	438,097	
23						

Selected explanations (concise)

- **VLOOKUP (B):** =VLOOKUP(A2,Sheet1!I2:J2539,2,false)
- Finds the matching Region value for State in Sheet1 range I2:J2539. Returns the Region (column 2). Used to populate Region names.
- **CONCATENATE(C):**
- =CONCATENATE(A2," : ",B2)
- Produces "State : Region" string for labels and grouping (used for reporting or labels in visuals).
- **COUNTIF (D):** =COUNTIF(Sheet1!I:I,Sheet1!I2)
- Counts occurrences of the state in column I — used to show number of orders/customers per state.
- **SUMIF (E):** =SUMIF(Sheet1!I:I,A2,Sheet1!P:P)
- Sums sales (Sheet1!P:P) for each state; drives Rank and contribution calculations.
- **RANK (F):** =RANK(E2,E\$2:E\$21,0)
- Ranks states by total sales (E) in descending order (0 = descending).

H	I	J	K	L	M	N	O
Corporate	Consumer	Home Office	Max Segment By Sale Qty	1st Order Date	Last Order Date	Days From Last Order Date	Last Order Format "MMM-YY"
259	406	160	Consumer	1/3/2014	26/12/2017	8 Years 6 Months 7 Day	Dec-17
150	380	124	Consumer	2/6/2014	15/12/2017	8 Years 6 Months 18 Day	Dec-17
51	118	51	Consumer	3/10/2014	29/12/2017	8 Years 6 Months 4 Day	Dec-17
56	95	25	Consumer	1/23/2014	30/12/2017	8 Years 6 Months 3 Day	Dec-17
27	47	41	Consumer	2/1/2014	17/12/2017	8 Years 6 Months 16 Day	Dec-17
342	540	253	Consumer	1/28/2014	29/12/2017	8 Years 6 Months 4 Day	Dec-17
156	255	116	Consumer	1/5/2014	18/12/2017	8 Years 6 Months 15 Day	Dec-17
138	219	102	Consumer	2/27/2014	26/12/2017	8 Years 6 Months 7 Day	Dec-17
42	47	4	Consumer	5/12/2014	11/12/2017	8 Years 6 Months 22 Day	Dec-17
48	30	4	Corporate	2/7/2014	25/12/2017	8 Years 6 Months 8 Day	Dec-17
140	164	68	Consumer	1/21/2014	22/12/2017	8 Years 6 Months 11 Day	Dec-17
84	176	17	Consumer	3/3/2014	18/12/2017	8 Years 6 Months 15 Day	Dec-17
39	160	35	Consumer	3/14/2014	17/12/2017	8 Years 6 Months 16 Day	Dec-17
64	84	14	Consumer	1/20/2014	28/12/2017	8 Years 6 Months 5 Day	Dec-17
52	84	23	Consumer	1/6/2014	26/11/2017	8 Years 7 Months 7 Day	Nov-17
30	47	5	Consumer	4/5/2014	21/12/2017	8 Years 6 Months 12 Day	Dec-17
554	996	417	Consumer	1/6/2014	29/12/2017	8 Years 6 Months 4 Day	Dec-17
159	239	93	Consumer	2/15/2014	28/12/2017	8 Years 6 Months 5 Day	Dec-17
80	108	44	Consumer	4/8/2014	22/12/2017	8 Years 6 Months 11 Day	Dec-17
51	65	20	Consumer	6/22/2014	30/12/2017	8 Years 6 Months 3 Day	Dec-17
							.

Formula explanations — continued (aggregation & segmentation)

- SUMIFS (H–J):**
=SUMIFS(Sheet1!Q:Q,Sheet1!I:I,A2,Sheet1!G:G,\$H\$1)
 - Aggregates segment-specific sale quantities (Sheet1!Q:Q) for the given state and for a specific segment value stored in \$H\$1/\$I\$1/\$J\$1 — used to determine Corporate/Consumer/Home Office contributions by state.
- XLOOKUP (K): =XLOOKUP(MAX(H2:J2),H2:J2,\$H\$1:\$J\$1)**
 - Finds the label of the segment (header in \$H\$1:\$J\$1) that corresponds to the maximum of H:J for each row — returns the Max Segment By Sale Qty.
- MINIFS / MAXIFS (L–M):**
=MINIFS(Sheet1!C:C,Sheet1!I:I,A2)
 - Finds earliest and latest order dates (Sheet1!C:C) for the state — used to compute recency and order age.
- DATEDIF + TEXT (N–O):**
**=DATEDIF(M2,today(),"Y")&" Years
 "&DATEDIF(M2,TODAY()),"YM")&" Months
 "&DATEDIF(M2,TODAY()),"MD")&" Day "**
 - Converts the max date into "Years Months Day" relative to TODAY and a short month code (e.g., Dec-17) for quick readability and timeline labels.

Formula explanations — financial and classification logic

Financial calculations and classification:

- **AVERAGEIF (Q):** =AVERAGEIF(Sheet1! :I,A2,Sheet1!P:P)
 - Calculates Avg Sale Per Customer by state from Sheet1 P:P for the state's rows.
- **SUMIF (R,S):** =SUMIF(Sheet1! :I,A2,Sheet1!S:S)
 - Aggregates profit or related totals by state (source: Sheet1 S:S). Both R and S use same formula in provided mandate — confirm mapping if R and S should be different source columns.
- **Profit Slab (T):**
=IF(R2<10000,"<10K",IF(and(R2>=10000,R2<=20000),"10-20K", ">20K"))
 - IF() categorizes total profit into <10K, 10-20K, or >20K. This slab is used to highlight low-profit states.
- **Conditional Formatting (U):** Instruction: highlight values <10K in red. In Excel: apply a rule with formula =R2<10000 and set font or fill red.

Q	R	S	T	U
Avg Sale Per Customer	Total Profit	Profit After Tax (18%)	Total Profit Slab (<10K,10-20K,>20K)	Highlight values <10K in red.
142	-3323.4274	c	<10K	-3323.4274
185	-6757.5101	No tax	<10K	-6757.5101
364	8186.0054	6,713	<10K	8186.0054
310	2338.2772	1,917	<10K	2338.2772
308	2081.7229	1,707	<10K	2081.7229
260	16252.9926	13,327	10-20K	16252.9926
171	-3587.3053	No tax	<10K	-3587.3053
162	-1821.7326	No tax	<10K	-1821.7326
176	331.4852	272	<10K	331.4852
146	1289.72	1,058	<10K	1289.72
182	554.883	455	<10K	554.883
256	-2889.9495	No tax	<10K	-2889.9495
174	2599.0921	2,131	<10K	2599.0921
119	-119.9297	No tax	<10K	-119.9297
128	1519.4214	1,246	<10K	1519.4214
216	1673.9977	1,373	<10K	1673.9977
186	15134.4495	12,410	10-20K	15134.4495
232	6090.7009	4,994	<10K	6090.7009
170	-1992.7181	No tax	<10K	-1992.7181
158	-27.641	No tax	<10K	-27.641
				37,533

Worked example — Texas (step-by-step)



State	Region	Total Order	Total Sale	Rank	Contribution	segments
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- VLOOKUP: returns the mapped state label for A2 from the lookup table.
 - CONCATENATE: "Texas : Central" — combines Region and State.
 - COUNTIF
→ 232 total orders (D).
 - SUMIF (Total Sale E) → 32,876.257 (E).
 - RANK → rank 3 by total sales.
 - Contribution = $32,876.257 / 438,097 = 7.50\%$.
 - Segment sums: Corporate 259, Consumer 406, Home Office 160 → Max Segment = Consumer.
 - First/Last order range: 1/3/2014 → 26/12/2017
→ Recency $\approx$ "8 Years 6 Months 7 Day".
 - Avg Sale per customer = 142.
- Total Profit = -3,323.43]
- Slab = "<10K" and highlighted red.

Analysis — Key findings (concise)

Top states by sales

California (22.11%), New York (17.31%), Texas (7.50%) — these three drive ~46.9% of total sales and should be prioritized for campaign investment.

Segment dominance

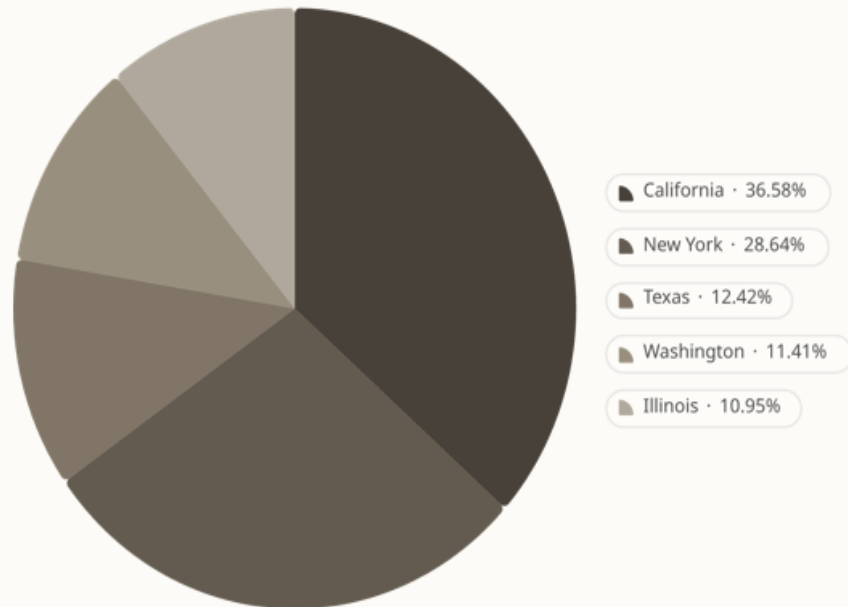
Consumer segment is the dominant contributor across most states; target consumer-focused promotions for higher yield.

Profit concerns

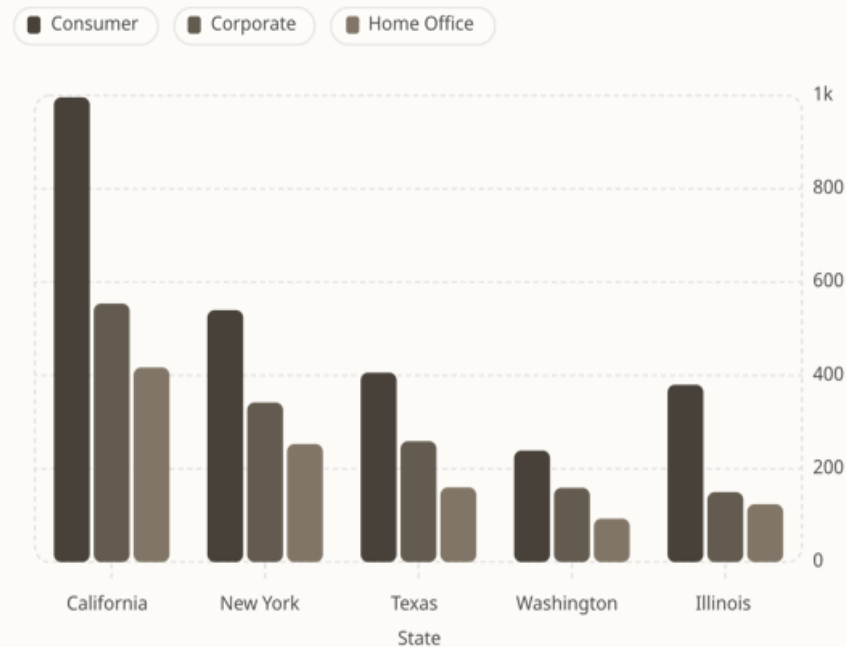
Many states have Total Profit <10K (highlighted red) — investigate pricing, returns, or cost allocation in those markets.



Visuals — Charts



Top 5 states contribution to total sales (slice labels show relative share).



Segment distribution (by sale quantity) for top states — highlights where Consumer segment volume is concentrated.

Practical notes — Implementation & verification

Steps to implement and validate formulas in your workbook:

- 1) Confirm Sheet1 ranges (I, P, Q, S) contain the expected fields (State, Sale, Qty, Profit).
- 2) Lock absolute references used in formulas (e.g., E\$2:E\$21, \$H\$1:\$J\$1) to avoid copy errors.
- 3) Validate XLOOKUP domain: ensure H2:J2 contain numeric comparators and \$H\$1:\$J\$1 headers are exact segment names.
- 4) Conditional formatting: create a rule to color profit cells red when < 10000 ($R2 < 10000$).
- 5) Test edge cases: missing dates (for DATEDIF), empty segment fields, or negative profits.

By Kshravya